Sarthak Kala

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Education

University of California San Diego — La Jolla, CA
M.S. Data Science (GPA: 3.9/4.0) — *Sep 2024 – Present (Expected Mar 2026)*
- **Coursework:** Safety in Generative AI; Deep Generative Models; Recommender Systems; ML Systems

Experience

Hewlett Packard Enterprise — Sunnyvale, CA
Machine Learning Engineering Intern — *Jun 2025 – Sep 2025*
- Built a **transformer-based root cause classifier** over switch log sequences, learning causal patterns to attribute network failures to config changes, hardware faults, or upstream issues, reducing engineer diagnosis time by 40%.
- Built a **multi-agent system** to parse switch documentation and auto-generate network test cases, reducing manual effort.

American Express — Gurugram, India
Senior Machine Learning Engineer — *Feb 2022 – Aug 2024*
- Built a **two-tower neural retrieval system for personalized ad ranking** serving 10M+ users, encoding transaction-derived user embeddings and multimodal creative features. Integrated **FAISS** for sub-10ms candidate retrieval from 10K+ creatives and trained an **XGBoost re-ranker** on 100M+ impressions incorporating real-time signals (frequency, budget pacing, intent), achieving 8% CTR lift and 12% ROAS improvement.
- Developed **CTR/CVR propensity models** for high-traffic referral surfaces processing 5M+ monthly sessions, using clickstream data. Enabled real-time widget personalization, increasing CTR by 20% and downstream conversions by 6%.
- Built **real-time personalized card ranking** for multi-card landing page, scoring user-card affinity using behavioral, contextual, and first-party signals. Deployed low-latency inference (sub-10ms) serving millions of daily visitors, increasing page CTR by 8% and application conversion by 3%.
- Designed and operated **large-scale feature pipelines** using **Spark** and Airflow, processing 100M+ daily events and powering real-time and batch features for ranking, propensity, and experimentation systems.
- Owned **end-to-end ML deployment infrastructure**, containerizing model services with Docker, orchestrating with Kubernetes, and implementing CI/CD pipelines with automated testing and rollback.

American Express — Gurugram, India
Machine Learning Engineer — *Jun 2019 – Jan 2022*
- Designed and deployed **hierarchical Bayesian ads measurement models (MMM)** with adstock and saturation to estimate incremental lift across different channels enabling data-driven budget reallocation and improving ROI by 12%.
- Designed and deployed **uplift modeling pipelines (X-Learner)** and validated impact through **A/B experiments**, increasing high-quality referrals by 18% without increasing incentive spend.
- Built a **customer social-influence graph** using referral links and co-spend proximity and trained **Node2Vec/GraphSAGE embeddings** to identify influential referrers, increasing new-referrer participation by 20%.
- Built a **production complaint segmentation system** using **Sentence-BERT embeddings** and unsupervised clustering to surface emerging customer pain-point themes, reducing manual review workload by 30%.

Selected Projects

Lost in Legal Language: AI Text Detectors on Legal Documents — UC San Diego
Evaluation of zero-shot AI text detectors under legal constraints — *2025*
- Demonstrated that **zero-shot AI text detectors do not generalize to legal documents**; even after domain fine-tuning, BiScope achieved only 66% recall and degraded sharply on 400–800 word texts.
- Constructed a 10K-document synthetic legal corpus across four LLMs and **re-implemented BiScope's bidirectional cross-entropy features and classifier** from scratch for controlled evaluation.

Memorization on Fine-Tuned LLMs (LoRA) — UC San Diego
Privacy and memorization analysis in parameter-efficient fine-tuning — *2025*
- Empirically showed that **LoRA fine-tuning induces severe privacy leakage**, with membership inference AUC reaching 0.95 by epoch 3 despite minimal verbatim copying.
- Designed controlled fine-tuning experiments sweeping epochs, rank, and data repetition, and evaluated entropy-adaptive mitigation objectives.

Technical Skills

Languages: Python, C/C++, Java, SQL, Shell
ML / Modeling: PyTorch, TensorFlow, JAX, Scikit-learn, Hugging Face; Bayesian modeling, causal inference
Infrastructure: Spark, Kafka, Airflow, Docker, Kubernetes, AWS, GCP; CI/CD, data pipelines
GenAI: LLMs, embeddings, RAG, FAISS; PEFT/LoRA; **LLM post-training** (instruction tuning, RLHF, continual learning); data-efficient learning